

## Quick Note on Wavenumber = Phase Constant = Propagation Constant

$$k = \omega \sqrt{\mu \epsilon} = \omega \sqrt{\mu_r \mu_0 \epsilon_r \epsilon_0} = \frac{\omega}{c} \sqrt{\mu_r \epsilon_r} = \frac{2\pi f}{c} \sqrt{\mu_r \epsilon_r} = \frac{2\pi}{\lambda_0} \sqrt{\mu_r \epsilon_r} = \frac{2\pi}{\lambda} \quad (\text{rad/m})$$

- $k$  = wavenumber of material
- amount of phase change a wave of frequency  $\omega$  will undergo per unit distance in the material

$\beta$  = propagating constant of EM wave, this is the constant at the phasor part governing phase change

- amount of phase change per unit distance propagated
- for a waveguide with the longitudinal direction along the  $z$ -axis
  - $\beta_z$  is the amount of phase change per unit distance propagated in the longitudinal direction
    - it describe the overall effect the waveguide has on the propagation
    - $\beta_z$  is the only “wanted” wavenumber as it governs the propagation in the direction the wave guide is guiding
    - for EE4112, the  $z$  is dropped to be just  $\beta$ , but sometimes it is not dropped
  - $k_c^2 = \beta_x^2 + \beta_y^2$ , describe how the wave is bouncing around when looking at a slice of waveguide
    - $k_c$  is the overall effect  $\beta_x$  and  $\beta_y$ , each describing how much the phase change when “step” in the direction of  $x$  and  $y$  (the transverse direction)

$$k^2 = \beta^2 + k_c^2$$

- the overall effect of the amount of phase change in the longitudinal and transverse direction in the waveguide must be the same amount of phase change of travelling in the same medium unbounded
- for a bouncing wave:
  - $k$  follows the bouncing
  - $\beta$  follows the longitudinal direction
  - $k_c$  follows the transverse direction
- The equation is called dispersion relation. Phase velocity in a waveguide depends on frequency; it is a dispersion device, not linear phase.

## Wave Properties (for any waveguide, any mode)

$$u_p = \text{phase velocity} = \frac{\omega}{\beta_z} = f\lambda_{\text{guided}}$$

$$\lambda_g = \text{guided wavelength} = \frac{2\pi}{\beta_z} \quad \lambda_c = \text{cutoff wavelength} = \frac{2\pi}{k_c}$$

$$k^2 = \beta^2 + k_c^2 \Leftrightarrow \frac{1}{\lambda^2} = \frac{1}{\lambda_g^2} + \frac{1}{\lambda_c^2}$$

$$Z = \text{wave impedance} = \frac{E_x}{H_y} = -\frac{E_y}{H_x}$$

## Wave Type Classification

## Transverse Field Components

TEM = Transverse ElectroMagnetic

$$E_z = 0 \quad H_z = 0 \quad k = \beta_z \quad k_c = 0$$

TM = Transverse Magnetic

$$E_z \neq 0 \quad H_z = 0 \quad k \neq \beta_z \quad k_c \neq 0$$

TE = Transverse Electric

$$E_z = 0 \quad H_z \neq 0 \quad k \neq \beta_z \quad k_c \neq 0$$

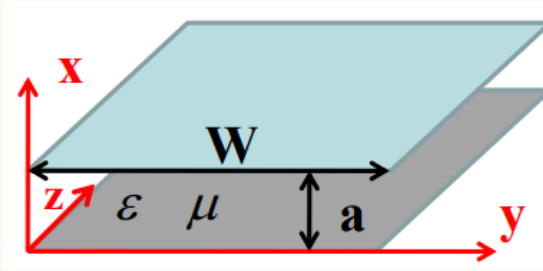
$$H_x = \frac{j}{k_c^2} \left( \omega \epsilon \frac{\partial E_z}{\partial y} - \beta \frac{\partial H_z}{\partial x} \right)$$

$$H_y = \frac{-j}{k_c^2} \left( \omega \epsilon \frac{\partial E_z}{\partial x} + \beta \frac{\partial H_z}{\partial y} \right)$$

$$E_x = \frac{-j}{k_c^2} \left( \beta \frac{\partial E_z}{\partial x} + \omega \mu \frac{\partial H_z}{\partial y} \right)$$

$$E_y = \frac{j}{k_c^2} \left( -\beta \frac{\partial E_z}{\partial y} + \omega \mu \frac{\partial H_z}{\partial x} \right)$$

## Parallel Plate Waveguide



TEM ( $E_z = 0, H_z = 0$ )

$$E = -\hat{x} \frac{V_0}{a} e^{-j\beta_z z} = -\hat{x} \frac{V_0}{a} e^{-jkz} \quad H = -\hat{y} \frac{V_0}{\eta a} e^{-j\beta_z z} = -\hat{y} \frac{V_0}{\eta a} e^{-jkz}$$

$V_0$  = potential at top plate with bottom grounded

$$u_{p(TEM)} = \frac{\omega}{k} = \frac{1}{\sqrt{\mu\epsilon}} = \frac{c}{\sqrt{\mu_r \epsilon_r}} = f\lambda \quad (m/s)$$

$$Z_{TEM} = \frac{E_x}{H_y} = \sqrt{\frac{\mu}{\epsilon}} = \eta_0 \sqrt{\frac{\mu_r}{\epsilon_r}} = \eta \quad (\Omega) \quad \eta_0 = \sqrt{\frac{\mu_0}{\epsilon_0}} = 120\pi = 377 \Omega$$

TM ( $E_z \neq 0, H_z = 0$ )

TE ( $E_z = 0, H_z \neq 0$ )

**ONLY TRUE FOR THIS XYZ CONFIG, IF OTHER CONFIG PROPAGATES THE OTHER WAY, WILL GET MINUS SIGN IN FRONT, SHOULD ALWAYS DO FROM MAXWELL**

$$H_y(x, z) = H_0 \cos(\beta_x x) e^{-j\beta_z z}$$

Points in the infinitely wide direction

$$E_x(x, z) = \frac{\beta_z}{\omega \epsilon} H_0 \cos(\beta_x x) e^{-j\beta_z z}$$

$$E_z(x, z) = \frac{j\beta_x}{\omega \epsilon} H_0 \sin(\beta_x x) e^{-j\beta_z z}$$

$$E_y(x, z) = E_0 \sin(\beta_x x) e^{-j\beta_z z}$$

Points in the infinitely wide direction

$$H_x(x, z) = -\frac{\beta_z}{\omega \mu} E_0 \sin(\beta_x x) e^{-j\beta_z z}$$

$$H_z(x, z) = \frac{j\beta_x}{\omega \mu} E_0 \cos(\beta_x x) e^{-j\beta_z z}$$

$$TM_m : m \in \{0 (TM_0 = TEM), 1, 2, \dots\}$$

$$TE_m : m \in \{1, 2, \dots\}$$

$$\beta_x = k_c = \frac{m\pi}{a}$$

$$\beta_z = \sqrt{k^2 - k_c^2} = \sqrt{k^2 - \beta_x^2} = \sqrt{(\omega \sqrt{\mu\epsilon})^2 - \left(\frac{m\pi}{a}\right)^2}$$

$$f_{c(TM_m)} = f_{c(TE_m)} = \frac{k_c}{2\pi\sqrt{\mu\epsilon}} = \frac{m}{2a\sqrt{\mu\epsilon}}$$

$f_{operating} > f_c$  to excite mode  $m$

$$\lambda_c = \frac{2\pi}{k_c} = \frac{2a}{m}$$

$$\lambda_g = \frac{2\pi}{\beta_z} = \frac{2\pi}{\sqrt{(\omega \sqrt{\mu\epsilon})^2 - \left(\frac{m\pi}{a}\right)^2}}$$

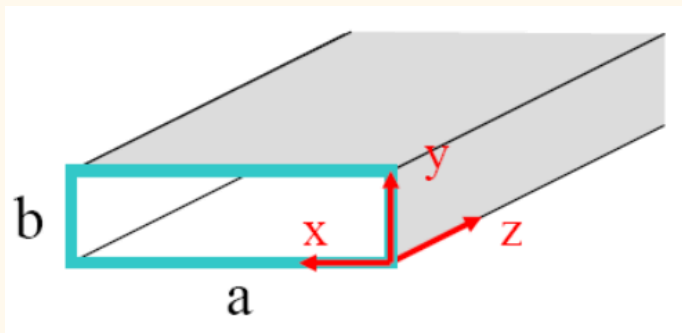
$$u_{phase} = \frac{\omega}{\beta_z} = \frac{\omega}{\sqrt{(\omega \sqrt{\mu\epsilon})^2 - \left(\frac{m\pi}{a}\right)^2}} = f\lambda_g \quad (m/s)$$

$$u_p u_g = c_{in\ medium}$$

$$Z_{TM} = \frac{E_x}{H_y} = \frac{\beta_z}{\omega\epsilon} = \frac{\beta_z}{k} \eta \quad (\Omega)$$

$$Z_{TE} = -\frac{E_y}{H_x} = \frac{\omega\mu}{\beta_z} = \frac{k}{\beta_z} \eta \quad (\Omega)$$

## Rectangular Waveguide



- there is no TEM, is due to TEM needing a static E field from potential difference
- **CONVENTION: a is long side, TM/TE<sub>m</sub>n, m for side a**

TM ( $E_z \neq 0, H_z = 0$ )

TE ( $E_z = 0, H_z \neq 0$ )

**ONLY TRUE FOR THIS XYZ CONFIG, IF OTHER CONFIG PROPAGATES THE OTHER WAY, WILL GET MINUS SIGN IN FRONT, SHOULD ALWAYS DO FROM MAXWELL**

$$E_z = E_0 \sin(\beta_x x) \sin(\beta_y y) e^{-j\beta_z z}$$

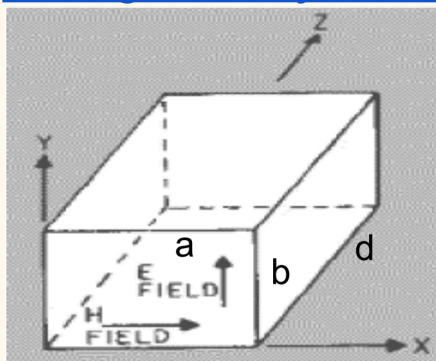
$$H_z = 0$$

$$H_z = H_0 \cos(\beta_x x) \cos(\beta_y y) e^{-j\beta_z z}$$

$$E_z = 0$$

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|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $E_x = \frac{-j\beta_z}{k_c^2} \beta_x E_0 \cos(\beta_x x) \sin(\beta_y y) e^{-j\beta_z z}$ $E_y = \frac{-j\beta_z}{k_c^2} \beta_y E_0 \sin(\beta_x x) \cos(\beta_y y) e^{-j\beta_z z}$ $H_x = \frac{j\omega\epsilon}{k_c^2} \beta_y E_0 \sin(\beta_x x) \cos(\beta_y y) e^{-j\beta_z z}$ $H_y = \frac{-j\omega\epsilon}{k_c^2} \beta_x E_0 \cos(\beta_x x) \sin(\beta_y y) e^{-j\beta_z z}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | $E_x = \frac{j\omega\mu}{k_c^2} \beta_y H_0 \cos(\beta_x x) \sin(\beta_y y) e^{-j\beta_z z}$ $E_y = \frac{-j\omega\mu}{k_c^2} \beta_x H_0 \sin(\beta_x x) \cos(\beta_y y) e^{-j\beta_z z}$ $H_x = \frac{j\beta_z}{k_c^2} \beta_x H_0 \sin(\beta_x x) \cos(\beta_y y) e^{-j\beta_z z}$ $H_y = \frac{j\beta_z}{k_c^2} \beta_y H_0 \cos(\beta_x x) \sin(\beta_y y) e^{-j\beta_z z}$ |
| $TM_{mn} : m, n \in \{1, 2, \dots\}$<br>Dominant Mode: $TM_{11}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | $TE_{mn} : m, n \in \{0, 1, 2, \dots\}$<br><b>WARN: only one of m or n can be 0</b><br>Dominant Mode: $TE_{10}$<br><b>WARN: a is long side, if b is long side then <math>TE_{01}</math></b>                                                                                                                                                                                      |
| $\beta_x = \frac{m\pi}{a} \quad \beta_y = \frac{n\pi}{b}$ $k_c^2 = \beta_x^2 + \beta_y^2 = \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2$ $\beta_z = \sqrt{k^2 - k_c^2} = \sqrt{(\omega\sqrt{\mu\epsilon})^2 - \left(\frac{m\pi}{a}\right)^2 - \left(\frac{n\pi}{b}\right)^2} = k\sqrt{1 - \left(\frac{f_c}{f}\right)^2}$ $f_{c(TM_{mn})} = f_{c(TE_{mn})} = \frac{k_c}{2\pi\sqrt{\mu\epsilon}} = \frac{1}{2\pi\sqrt{\mu\epsilon}} \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2} = \frac{c}{2\sqrt{\mu_r\epsilon_r}} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$ $f_{operating} > f_c \text{ to excite mode } mn$ $\lambda_c = \frac{2\pi}{k_c} = \frac{2\pi}{\sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2}} = \frac{2}{\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}}$ $\lambda_g = \frac{2\pi}{\beta_z} = \frac{2\pi}{\sqrt{(\omega\sqrt{\mu\epsilon})^2 - \left(\frac{m\pi}{a}\right)^2 - \left(\frac{n\pi}{b}\right)^2}} = \frac{2\pi}{k\sqrt{1 - \left(\frac{f_c}{f}\right)^2}} = \frac{\lambda}{\sqrt{1 - \left(\frac{f_c}{f}\right)^2}}$ $u_{phase} = \frac{\omega}{\beta_z} = \frac{v}{\sqrt{1 - \left(\frac{f_c}{f}\right)^2}} = f\lambda_g \quad (m/s)$ $v = \frac{1}{\sqrt{\mu\epsilon}} = u_p \text{ not in waveguide but in same material}$ $u_{group} = v\sqrt{1 - \left(\frac{f_c}{f}\right)^2} \quad (m/s)$ |                                                                                                                                                                                                                                                                                                                                                                                  |
| $Z_{TM} = \frac{E_x}{H_y} = \frac{\beta_z}{\omega\epsilon} = \frac{\beta_z}{k} \eta \quad (\Omega)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $Z_{TE} = \frac{E_x}{H_y} = \frac{\omega\mu}{\beta_z} = \frac{k}{\beta_z} \eta \quad (\Omega)$                                                                                                                                                                                                                                                                                   |

## Rectangular Cavity Resonators



- **CONVENTION:** a is long side, TM/TE<sub>mnp</sub>, m for side a

**TM** ( $E_z \neq 0, H_z = 0$ )

$$E_z = E_0 \sin(\beta_x x) \sin(\beta_y y) \cos(\beta_z z)$$

$$H_z = 0$$

$$E_x = \frac{-j\beta_z}{k_c^2} \beta_x E_0 \cos(\beta_x x) \sin(\beta_y y) \cos(\beta_z z)$$

$$E_y = \frac{-j\beta_z}{k_c^2} \beta_y E_0 \sin(\beta_x x) \cos(\beta_y y) \cos(\beta_z z)$$

$$H_x = \frac{j\omega\epsilon}{k_c^2} \beta_y E_0 \sin(\beta_x x) \cos(\beta_y y) \cos(\beta_z z)$$

$$H_y = \frac{-j\omega\epsilon}{k_c^2} \beta_x E_0 \cos(\beta_x x) \sin(\beta_y y) \cos(\beta_z z)$$

**TE** ( $E_z = 0, H_z \neq 0$ )

$$H_z = H_0 \cos(\beta_x x) \cos(\beta_y y) \sin(\beta_z z)$$

$$E_z = 0$$

$$E_x = \frac{j\omega\mu}{k_c^2} \beta_y H_0 \cos(\beta_x x) \sin(\beta_y y) \sin(\beta_z z)$$

$$E_y = \frac{-j\omega\mu}{k_c^2} \beta_x H_0 \sin(\beta_x x) \cos(\beta_y y) \sin(\beta_z z)$$

$$H_x = \frac{j\beta_z}{k_c^2} \beta_x H_0 \sin(\beta_x x) \cos(\beta_y y) \sin(\beta_z z)$$

$$H_y = \frac{j\beta_z}{k_c^2} \beta_y H_0 \cos(\beta_x x) \sin(\beta_y y) \sin(\beta_z z)$$

**TM:**  $m, n \in \{1, 2, \dots\}, p \in \{0, 1, 2, \dots\}$

**Dominant Mode:**  $TM_{110}$

**TE:**  $m, n \in \{0, 1, 2, \dots\}, p \in \{1, 2, \dots\}$

**WARN:** only one of m or n can be 0

**Dominant Mode:**  $TE_{101}$

**WARN:** a is long side, if b is long side then  $TE_{011}$

$$\beta_x = \frac{m\pi}{a}$$

$$\beta_y = \frac{n\pi}{b}$$

$$\beta_z = \frac{p\pi}{d}$$

$$f_{TM_{mnp}} = f_{TE_{mnp}} = \frac{c}{2\sqrt{\mu_r \epsilon_r}} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2 + \left(\frac{p}{d}\right)^2}$$

$$Q = \frac{f_{mnp}}{\Delta f_{3dB}}$$